

Mirra / QS Tools — Full Completion Brief

The Completed Closed-Loop Business Recovery Tool

1. Executive Summary

Mirra is a closed-loop business recovery system.

On completion, it will help a business owner understand whether their business model is working, whether their quotes are safe, whether their jobs are proving the model, and whether monthly financial performance is improving or drifting away from target.

Mirra will connect:

```
graph TD; A[Accounting truth] --> B[Business recovery model]; B --> C[Required rates]; C --> D[Quote checking]; D --> E[Works orders]; E --> F[Purchase orders]; F --> G[Labour / time tracking]; G --> H[Asset usage]; H --> I[Supplier invoices]; I --> J[Job actuals]; J --> K[Reconciliation]; K --> L[Xero]; L --> M[Monthly P&L]; M --> N[Model feedback]; N --> O[Approved model improvement]; O --> P[Future quote targets];
```

Accounting truth
↓
Business recovery model
↓
Required rates
↓
Quote checking
↓
Works orders
↓
Purchase orders
↓
Labour / time tracking
↓
Asset usage
↓
Supplier invoices
↓
Job actuals
↓
Reconciliation
↓
Xero
↓
Monthly P&L
↓
Model feedback
↓
Approved model improvement
↓
Future quote targets

This loop is the product.

Mirra is not just a dashboard, quoting tool, job costing tool, estimating tool, accounting add-on, or project management system.

It is a business model control system.

It turns accounting data into recovery targets, turns those targets into rates and quote rules, checks quotes before work starts, tracks actual job delivery, reconciles job reality back to accounting, and uses the next P&L to prove whether the model is working.

The current interface is complex because the engine is being built in full. The completed user interface will be much simpler.

The final user experience should follow this principle:

Answer first.
Reason second.
Evidence third.
Machinery only on drilldown.

The system underneath can remain complex because that complexity is what creates trusted answers.

The user should see the simple truth first.

2. Core Product Statement

Mirra answers:

Is the business model working?
Are quotes aligned to that model?
Are jobs proving or disproving the model?
Is monthly P&L performance improving?
What needs to change next?

Most systems answer only part of this:

What does the P&L say?
Did this job make money?
What is the quote value?
What are the job actuals?
What is the current margin?

Mirra asks the deeper question:

Does every job, quote, invoice, labour hour, asset hour, supplier cost, and monthly P&L result align with the business recovery model?

That is the completed product.

3. The Full Closed Loop

The completed Mirra loop is:

```
Monthly P&L
↓
Business recovery model
↓
Required labour / asset / overhead / COG recovery
↓
Rate targets
↓
Business Modelling target
↓
Quote Checker
↓
Approved quote
↓
Works order
↓
Purchase orders
↓
Labour / time tracking
↓
Asset usage tracking
↓
Supplier invoices
↓
Job actuals
↓
Job reconciliation
↓
Xero
↓
Next monthly P&L
↓
Model comparison
↓
Model feedback
↓
Model review / approval
```

↓
Updated target model
↓
Future quote targets

This means the system works in two connected lenses.

Job lens

Did this quote recover the model?
Did this works order follow the quote?
Did POs stay inside the allowance?
Did labour stay inside allowed hours?
Did asset usage recover asset cost?
Did supplier invoices match the expected cost?
Did the job prove the model?

Business lens

Did the whole business recover as expected this month?
Did the P&L prove the model?
Did labour recover?
Did assets recover?
Did overheads recover?
Did material margin hold?
Did rates need to change?

Both lenses use the same recovery logic.

4. What Mirra Will Do on Completion

4.1 Import and preserve accounting truth

Mirra will import accounting data from Xero, Excel, CSV, manual import, or future API sources.

It will preserve:

- original account names
- original supplier names
- source file metadata
- source period
- signed values
- GST/tax where relevant
- raw and normalised names
- source trace

- import history

Xero remains the accounting source of truth.

Mirra does not replace Xero.

Mirra uses Xero and P&L data to build and test the business model.

4.2 Build the business operating model from the P&L

Mirra will use the P&L as the accounting anchor.

It will classify and analyse:

- revenue
- COGS / direct costs
- labour
- assets
- general overheads
- finance / interest
- other income
- net profit
- accounting benchmark totals

It will then bridge the P&L to Mirra operating truth.

The user should be able to see:

What the P&L says
What Mirra says
Why they differ
Which module caused the difference
Whether the difference is classification, timing, accounting treatment, or
operating model treatment

The system will not hide differences.

It will explain them.

4.3 Separate macro business baseline from micro recovery strategy

Mirra will clearly separate the business operating baseline from the recovery strategy.

Macro layer

The macro layer answers:

What does the business cost to be open?
Does the trading margin pool cover the business cost?
What does the business cost per open hour?

It uses business opening hours and total operating cost.

Micro layer

The micro layer answers:

How will the business recover that cost?
Which driver carries the burden?
How do labour, assets, materials, overheads, units, and operating groups recover cost?

It may use:

- productive labour hours
- selected recovery hours
- asset utilisation hours
- unit drivers
- group drivers
- product drivers

This prevents the system from confusing:

business availability
productive labour capacity
selected recovery strategy
asset utilisation
unit recovery
cost allocation drivers

The completed UI must label these clearly.

4.4 Show the current business mirror

Business Summary will show the current operating position.

It will answer:

What is the business doing now?
What margin does it generate?
What cost burden must that margin carry?
What is left after labour, assets, and overheads?
Is the current business position positive, negative, or under pressure?

The core chain is:

Revenue
- COGS
= Margin Pool

Margin Pool
- Labour
= Residual Margin after Labour

Residual Margin after Labour
- Assets
- Overheads
= Net Position

A negative result is not a system failure.

A bad business result is not the same as bad data.

4.5 Stress-test the first failure point: labour

Revenue Reality will test:

Does margin after COGS survive labour?

It isolates:

Revenue
- COGS
= Margin Pool

Margin Pool
- Labour
= Margin After Labour

This tells the user whether the business is already under pressure before assets and overheads are considered.

This will become one of the simplest and most powerful user-facing outputs:

Labour is covered by margin.

or:

Labour consumes more than the margin pool.

4.6 Stress-test productive assets

Asset Reality will test whether productive assets justify themselves at their entered utilisation.

It will answer:

Are productive assets recovering themselves?
What does each asset need to earn per hour?
Are assets underutilised?
Which asset creates the highest recovery pressure?

Example:

Digger annual cost
÷ Digger annual utilisation hours
= Required digger recovery rate

This prevents asset cost from being hidden inside general business cost.

4.7 Build the operating structure

Cost Allocation will define how the business is operationally organised.

The completed structure will be:

Division
↓
Operating Group
↓
Productive Labour Groups
Productive Assets
Automatic Overhead

Cost Allocation will answer:

Which divisions exist?
Which operating groups belong to each division?
Which labour groups are assigned?
Which productive assets are assigned?
How is overhead distributed?
Is any cost unassigned?
Has anything been over-assigned?
Is the structure valid enough to support recovery?

It will not build rates.

It will not make the final business decision.

It only defines the operating structure.

4.8 Test whether the structure recovers

Recovery Summary will automatically test whether the allocated structure can recover its cost burden.

It will answer:

Does this structure recover?
Which group is under-recovered?
What is the required recovery amount?
What is the recovery shortfall?
What recovery driver is being used?

Recovery Summary will be a zero-input page.

The user should not choose the recovery method there.

The recovery method is determined by the business mode and upstream structure.

Recovery Summary does not create sell rates.

It hands the recovery requirement to Rate Builder.

4.9 Build practical rates

Rate Builder will answer:

What do I need to charge?

It will build:

- labour rates
- machine rates
- machine + operator rates
- crew rates
- product/unit rates where valid
- minimum recoverable rates
- fully burdened cost rates
- suggested sell rates
- current rate comparison
- target rate comparison
- margin / markup options

Rate Builder converts the model into chargeable rates.

It does not create source costs.

It does not change the P&L.

It does not issue the final business verdict.

4.10 Diagnose the business outcome

Business Outcome will answer:

What is happening?
Why is it happening?
Where is the pressure coming from?
What needs to change?

It will classify pressure such as:

- revenue shortfall
- COG pressure
- labour pressure
- asset pressure
- overhead pressure
- recovery gap
- rate under-recovery
- mixed pressure
- source data blocked

Business Outcome is the diagnosis layer.

It converts technical recovery results into a plain-English business interpretation.

4.11 Model the selected fix

Business Modelling will sit between Business Outcome and Quote Checker.

The relationship is:

```
Business Outcome = this is the problem
Business Modelling = this is the selected fix
Quote Checker = does this quote follow the selected fix?
```

Business Modelling will allow the user to model:

- supplier cost changes
- category cost changes
- total COG changes
- labour productivity changes
- labour cost changes
- asset utilisation changes
- asset cost changes
- overhead changes
- selected rate changes
- all-rate changes
- margin target changes
- combined scenarios

It will create the selected target model and quote checker target set.

Quote Checker will only enforce selected targets.

No unselected scenario should flow silently into quote checking.

4.12 Check quotes before work starts

Quote Checker will enforce the selected model.

It will answer:

```
Does this quote align with the business model?
Does it recover minimum cost?
Does it meet the target margin?
Which lines are under-recovering?
Which lines miss the target rate?
Does the full quote pass or fail?
```

Quote Checker will compare quote lines against:

- approved minimum recoverable rates
- approved suggested sell rates
- selected Business Modelling targets
- selected margin targets
- selected markup targets
- material assumptions
- labour assumptions
- asset assumptions
- overhead assumptions

Quote Checker does not decide the model.

It checks whether the quote follows the approved model.

5. Practical Delivery Layer

The completed system must not stop at quote checking.

It must turn an approved quote into live operational control.

5.1 Approved quote to live job

Once a quote is accepted, Mirra should create a live job.

The live job should carry forward:

- job number
- job name
- client
- quote version
- approved quote value
- approved model version
- approved target set
- approved labour allowance
- approved material allowance
- approved asset allowance
- approved overhead recovery target
- Quote Checker result

The live job becomes the bridge between the model and actual delivery.

5.2 Works order capability

From the live job, Mirra should create works orders.

A works order should tell the business:

This is the job we priced.
This is the quote version we approved.
This is the model it must follow.
These are the hours allowed.
These are the material allowances.
These are the asset / machine assumptions.
These are the recovery targets.

The works order is the operational instruction.

It should be connected to:

- quote line
- SOQ line
- task
- labour allowance
- asset allowance
- material allowance
- recovery target

5.3 Purchase order capability

Mirra should create or track purchase orders against the job.

Purchase orders should link to:

- job
- quote line
- works order
- SOQ line
- supplier
- material category
- expected quantity
- expected unit cost
- expected total cost
- actual supplier invoice cost

This allows Mirra to compare:

Quoted allowance
vs
Purchase order commitment
vs
Supplier invoice

vs
Actual job result

Purchase orders become the control point between the quote and actual material/subcontractor cost.

5.4 Labour app / time tracking capability

Mirra should include a labour tracking app or time management layer.

This layer should record:

- live job
- staff member
- staff type
- work date
- start time
- finish time
- break
- task
- work category
- notes

Initial work categories should include:

- productive
- travel
- setup
- admin
- delay
- rework
- variation
- other

The labour app should compare:

Quoted labour hours allowed
vs
Actual productive labour hours

And:

Quoted labour cost
vs
Actual labour cost

This will show whether a job failed because:

- the quote allowed too few hours
 - the crew took too long
 - non-productive time was too high
 - rework occurred
 - variation time was not recovered
 - labour assumptions were wrong
-

5.5 Asset usage tracking

Where assets are material to recovery, Mirra should track asset usage against jobs.

Asset usage should connect:

- asset
- live job
- quote line or works order
- usage hours
- recovery rate
- recovered value
- actual cost basis

This will show whether productive assets recovered their required cost in real work.

5.6 Supplier invoices and actual costs

Supplier invoices should be matched to:

- job
- PO
- quote line
- SOQ line
- supplier
- material category
- cost type

The system should compare:

Expected PO cost
vs
Actual invoice cost

And:

Quoted allowance
vs
Actual supplier cost

This gives Mirra the evidence needed to identify supplier pressure, material pressure, cost overruns, and margin compression.

5.7 Job reconciliation

At job reconciliation, Mirra should compare:

Quote
vs
Works order
vs
Purchase orders
vs
Supplier invoices
vs
Labour time
vs
Asset usage
vs
Actual job result

It should answer:

- Did the quote recover the required cost?
- Did purchase orders stay inside allowance?
- Did invoices match POs?
- Did labour stay inside allowed hours?
- Did asset usage recover asset cost?
- Did the job meet margin target?
- Did the job support the business recovery model?

This is stronger than normal job costing.

Normal job costing asks:

Did the job make money?

Mirra asks:

Did the job prove the business recovery model?

6. Return to Xero and P&L Loop

Once job actuals, supplier invoices, labour, asset usage, and invoicing are reconciled, the financial result should flow back to Xero.

Xero remains the accounting record.

Mirra remains the business model control system.

The intended relationship is:

```
Mirra checks and controls the model.  
Xero records the financial truth.
```

The completed loop is:

```
Mirra checks the quote before work starts.  
Mirra creates works orders and POs.  
Mirra tracks labour, assets, supplier invoices, and job actuals.  
Actual financial results are reconciled and recorded back into Xero.  
Xero produces the monthly P&L.  
Mirra imports the monthly P&L.  
Mirra compares actual P&L performance back to the model.  
Mirra creates model feedback.  
The user approves or rejects model improvements.  
Future quote targets are updated only after approval.
```

This is the recurring loop that validates the product.

7. Model Feedback and Learning

The Model Feedback Loop will turn actual job performance into structured review signals.

It will identify:

- rates that may need review
- supplier pressure
- material category pressure
- labour productivity problems
- asset under-recovery
- overhead under-recovery
- quote alignment failures
- repeated issues
- positive signals where the model worked

It will distinguish between:

```
quote failure
execution failure
supplier variance
labour productivity variance
asset variance
scope variance
data quality issue
one-off exception
repeated issue
```

Feedback is evidence.

It does not automatically change the model.

8. Model Review and Approval

Model Review / Approval will control whether feedback becomes a model change.

No model value should change silently.

The user will be able to:

- approve rate changes
- reject one-off issues
- defer review items
- request more data
- approve new target models
- approve quote checker target sets
- preserve version history
- roll back if needed

The core rule is:

```
Feedback is evidence.
Approval is decision.
```

This means Mirra can learn from actual performance without corrupting approved rates or assumptions automatically.

9. Data Quality and Trust

The Data Quality Dashboard will protect trust.

It will separate data problems from valid commercial results.

Data problems include:

- missing P&L
- missing revenue
- missing COG mapping
- missing labour hours
- missing asset utilisation
- failed reconciliation
- duplicate cost
- unmatched quote line
- missing PO link
- missing invoice link
- missing source trace

Commercial results include:

- low margin
- labour consumes margin
- high asset recovery pressure
- rate under-recovery
- quote margin shortfall
- job loss
- supplier cost pressure

Commercial failure should not automatically block the model if source data is trusted.

A bad business result is not the same as bad data.

10. Reporting and Analytics

Reporting / Analytics will show historical and model performance.

It will report on:

- business snapshot
- revenue trend
- COG trend
- margin trend
- labour trend
- asset trend
- overhead trend
- recovery gap trend
- quote performance
- job profitability
- supplier pressure
- COG category pressure

- rate performance
- model version performance
- data quality trend

It will support Power BI and external reporting through fact and dimension tables.

Reporting shows evidence.

It does not edit the model.

11. Storage, History, and Source Trace

The Database / Storage / Export Layer will preserve Mirra's memory.

It will store:

- source imports
- cleaned source rows
- module outputs
- approved model versions
- approved rate sets
- quote checker target sets
- quote checks
- quote check lines
- job records
- SOQ rows
- PO rows
- job actual costs
- job profitability
- feedback signals
- approval audit trail
- reporting fact tables
- reporting dimension tables

The database stores.

Modules calculate.

Exports publish.

Reporting consumes.

No approved model value should exist without source trace.

No job actual cost should exist without source trace.

No quote checker target should exist without source trace.

12. Integrations and External Systems

Mirra will integrate with external systems carefully.

The first major integration is Xero.

Future integrations may include:

- payroll
- timesheets
- quote systems
- PO systems
- job management systems
- Procore or similar tools

The Integrations / API Layer will:

- import data
- normalise data
- deduplicate imports
- preserve raw source trace
- preserve GST/tax
- preserve signed values
- preserve credit notes
- show sync status
- show stale data warnings
- hand clean rows to source modules

It will not:

- decide business outcomes
- build rates
- check quotes
- approve models
- silently change mappings

13. Procore / Job-System Relationship

Mirra should not replace Procore or similar systems.

The rule is:

Do not replace Procore.
Do not rebuild Procore.
Import only the economic signal needed to test the model.

Useful imported data may include:

- labour hours
- job costs
- asset usage
- commitments
- variations
- purchase orders
- job actuals

The relationship is:

```
Xero = financial truth
Procore / job systems = operational activity
Mirra = business model control
```

Mirra sits between financial truth and operational activity to test whether the business recovery model is working.

14. Final User Interface Direction

The current UI is complex because the full engine is still visible.

That is acceptable during build.

On completion, the UI should be much simpler.

The final user should not need to understand:

- every module
- every variable
- every calculation contract
- every warning object
- every trust state
- every source trace object
- every downstream handoff

The user should first see:

```
Where the business should be
Where the business actually is
The gap
The cause
The next action
```

The interface should be simplified into:

- dashboard cards
- plain-English explanations
- traffic-light status
- guided setup
- clear warnings
- drilldown panels
- source proof where needed
- mobile-friendly workflows

The final UX principle is:

Answer first.
Reason second.
Evidence third.
Machinery only on drilldown.

15. The Simplified Front Door

The first screen should answer:

Is the business model working?
Are we above or below baseline?
What is the main pressure point?
What should we fix first?
Are our quotes following the model?
Are jobs proving the model?
Is monthly performance improving or drifting?

Suggested dashboard outputs:

- Business recovery status
- Actual vs baseline performance
- Main leakage area
- Required rate warning
- Quote pass rate
- Job performance trend
- Data trust status
- Recommended next action

The user should not start inside Cost Summary, Recovery Summary, Cost Allocation, or Rate Builder.

Those are engine rooms.

The front door should show the answer.

16. Baseline vs Actual View

A critical completed feature is baseline vs actual.

The user must be able to see:

Expected model performance
vs
Actual business performance

This should include:

- baseline line
- actual line
- gap
- monthly drift
- cumulative drift
- pressure source
- recommended action

A simple performance line is not enough.

The system must compare performance to the selected model.

That is how the user knows whether the business is improving or drifting.

17. Plain-English Output Examples

The completed system should be able to tell the user:

Your business is below the recovery baseline.

Labour is consuming more margin than the model allows.

Your digger requires \$X/hr to justify its current utilisation.

This quote is \$Y below the minimum recoverable rate.

This quote covers cost but misses the target margin.

Concrete supplier movement is reducing margin by \$Z.

This job passed the quote check but failed in actual delivery.

The same labour productivity issue has appeared on three jobs.

The model should be reviewed before future quotes use the same rate.

Your data is blocked because asset utilisation is missing.

This is a commercial loss, not a data failure.

The user should not need to interpret technical tables before seeing these answers.

18. What Makes Mirra Different

Mirra is different because it connects the whole business loop.

It does not only ask:

Did this job make money?

It asks:

Did this job prove the recovery model?

It does not only ask:

What does the P&L show?

It asks:

Did the monthly P&L prove that the recovery model is improving or failing?

It does not only ask:

What rate should I charge?

It asks:

Does that rate recover labour, assets, overheads, materials, and the required business outcome?

This is the strategic difference.

19. AI Positioning

AI can explain the result, summarise the issue, and help the user understand the next action.

But AI needs structured data and reliable logic.

Mirra provides:

- source truth
- recovery model
- approved targets
- quote checks
- job actuals
- source trace
- feedback rules
- model approval history

AI can sit on top of Mirra as a future explanation layer.

Positioning:

AI can explain the answer.
Mirra creates the answer.

or:

Mirra is the structured recovery engine.
AI is the future interface layer.

20. First Sellable Product Test

The first sellable version does not need every future layer.

The first sellable test should prove:

A small business owner can see, in plain English, whether their rates and quotes are recovering the actual monthly cost of running the business.

The first sellable loop should be:

Monthly P&L
↓
Recovery model

↓
Required rates
↓
Quote check
↓
Monthly result comparison

Then the next layers add:

- works orders
- purchase orders
- labour app / time tracking
- job actuals
- Xero reconciliation
- model feedback
- model approval
- Power BI reporting
- deeper integrations

The first version must show value quickly.

21. Complete Capability Summary

On completion, Mirra will be able to:

1. Import accounting data from Xero or files.
2. Preserve P&L source truth and source trace.
3. Classify revenue, COGS, labour, assets, and overheads.
4. Build the real business cost burden.
5. Calculate operating cost per open business hour.
6. Show whether trading margin covers the operating baseline.
7. Stress-test whether margin survives labour.
8. Stress-test whether productive assets justify themselves.
9. Build divisions and operating groups.
10. Assign productive labour and productive assets to groups.
11. Automatically distribute overheads.
12. Test whether the operating structure recovers cost.
13. Build minimum recoverable rates and suggested sell rates.
14. Diagnose the business outcome.
15. Model selected fixes and target assumptions.
16. Create quote checker target sets.
17. Check quotes against approved targets.
18. Flag under-recovery and margin shortfall.
19. Turn approved quotes into live jobs.
20. Create works orders from accepted quotes.
21. Create and track purchase orders.
22. Track labour time against live jobs.
23. Track productive, travel, setup, admin, delay, rework, and variation time.
24. Track asset usage against jobs.

25. Match supplier invoices to POs and job lines.
 26. Reconcile actual job costs against the quote and model.
 27. Return reconciled financial reality to Xero.
 28. Import the next monthly P&L from Xero.
 29. Compare actual P&L performance against the model.
 30. Create feedback signals from actual performance.
 31. Require approval before changing model assumptions.
 32. Store approved model versions and quote targets.
 33. Preserve audit history and rollback capability.
 34. Separate data quality issues from business performance issues.
 35. Report trends across quotes, jobs, rates, suppliers, labour, assets, and model versions.
 36. Export reporting-ready data to Power BI or other systems.
 37. Integrate with future job systems without replacing them.
 38. Show the user simple answers while preserving evidence underneath.
-

22. Final Product Promise

The completed Mirra system will be able to say:

This is what your business costs.

This is what it needs to recover.

This is what your current margin supports.

This is where the recovery gap is.

These are the rates you need.

This quote does or does not meet the model.

This works order follows the approved quote.

This purchase order is inside or outside the allowance.

This labour time is inside or outside the allowed hours.

This supplier invoice matches or exceeds the PO.

This job did or did not prove the model.

This month did or did not improve the business.

This is what needs to change next.

That is the completed tool.

23. Final Principle

Mirra is a closed-loop recovery system.

It uses accounting truth to build the model.

It uses the model to set rates and quote targets.

It checks quotes before work starts.

It turns approved quotes into works orders, purchase orders, labour tracking, and job control.

It reconciles actual job performance back to Xero.

It uses the next monthly P&L to prove whether the model worked.

It creates feedback for controlled model improvement.

The current system is complex because the engine is being built.

The completed product must feel simple because the user should only see the answer first.

Final rule:

Build the complex engine.

Show the simple truth.

Let the user drill down only when they need proof.